

Git Documentation & Organization

Goal: The goal of the document is to provide information on the files available in Github, as well as those we could not make publicly available, so that a user could understand what information they would need to compile to run the model or apply it elsewhere.

Notes: If a name of a file is given, it is included in the Github repository. If the name is not given and the contents/information from it are included, then it is not publicly available.

Subfolder: data

costs

- File: Csv file with cost values for deployment order in linear programming
 - Columns: Category (name of source), Cost (values can be negative or positive)

data_system

- File: Csv file with bathymetry information for Loch Lomond
 - Columns: Index, Stage (ft NCDD), Stage (ft NAVD88), Storage (acre-ft), Storage (Mgal), Area (ac)

dcc_data

- File: Csv file with DCC model coefficients
 - Name: DCCcoefs.csv
- File: Csv file with single-family households data for use in simulation model
 - Columns: account, pipe size (in), pool, bathrooms, main area (sqft), tax value, census tract/block group, map_inc_1 ... map_inc_10, mean residuals (real)

Input_climate_stress_test

- Subfolder: FLOW
 - Files: Csv files for each stochastic simulation
 - Columns: Date; Streams (raw): Bigtrees, Laguna, Liddel, Majors, Newell, Taitside; Streams (Na): Bigtrees, Laguna, Liddel, Majors, Taitside; is it a critically dry year? (binary); first flush (binary)
- Subfolder: WEATHER
 - Files: Csv files for each stochastic simulation
 - Columns: Date, Precipitation, Evaporation, Turbidity (San Lorenzo River), Turbidity (North Coast), Turbidity (North Coast Majors)
 - Note: Turbidity is binary

climate_monthly.zip

- Contains: files for each stochastic climate scenario with monthly precipitation, temperature, and evapotranspiration data for us in the econometric model

Subfolder: model_assumptions_and_scenarios

- File: JSON file with assumptions related to the cashflow model
 - Name: cashflow_rate_assumptions.json
- File: JSON file with assumptions related to the infrastructure financing model, including infrastructure options (costs, deployment time, etc.)
 - Name: inf_planning_assumptions.json
- File: Csv file with water resources model assumptions involving single values
 - Columns: Assumption (name of assumption), SCWSM-Option_Analysis... (version name, most are based on the demand scenario, examples include: SCWSM-Demand_Baseline)
- File: Excel file with water resources model assumptions requiring monthly profiles
 - Columns: Assumption (name); 1, 2, ... (month of year)
 - Tabs: Names for each version, i.e., SCWSM-Option_Analysis
- Files created in simulation process:
 - used_assumptions_PROFILES.csv: contains the monthly profiles utilized in the model
 - used_assumptions_CSTE.csv: contains the single values utilized in the model
- File: JSON file with sensitivity analysis-relevant parameters
 - Name (example): file_SA.json
 - Notes: The code has been designed to be able to input an empty name for this file or a file with any combination of the possible sensitivity analysis parameters. In this example, all filenames are the baseline files, except resampled_households.csv, which is not included for data privacy reasons.
- File: Csv file containing the values for the hydrological condition type by month of year
 - Columns: Month, Driest (5), Dry (4), Normal (3), Wet (2), Wettest (1)

Subfolder: model > SCWSM-Option_Analysis > Parent

- File: JSON file containing the parameters used to build the Pywr model
 - Name: CST_SCWSM_OPTION_ANALYSIS_SA.json
 - Notes: Parts of this file have been removed including all nodes, all edges, some parameters, and some recorders
- Note: create subfolder to save individual json files into for stochastic simulations

Subfolder: scripts

- File: Python file that sets up the Pywr model for that specific simulation, including importing the climate data and the sensitivity analysis parameters
 - Name: Setup_SCWSM_Option_Analysis_CST.py
- File: Python file that defines the class householdDemand
 - Name: household_demands.py
- File: Python file that contains all of the parameter classes input into Pywr built for this project and paper
 - Name: scwsm_demands.py
- File: Python file with the parameter classes for the alternative water supply options and operations
 - Name: scwsm_alternative_supply_option_operations.py
 - Parameter classes for specific water resources system removed, including:
 - SUPPLY_GAP: specific to SCWD
 - FELTON_DIV_TO_LL: specific to SCWD
 - DESALINATION_PLANT_YIELD/DESALINATION_PLANT_YIELD_ROF,: determines the desalination plant yield based on the month of the year and storage levels
 - DIRECT_POTABLE_REUSE_PLANT_YIELD: determines the DPR plant yield based on the month of the year and storage levels
 - DEMAND_AFTER_RATIONING: determines the value of reduced urban demand depending on if rationing/curtailment is implemented in the model
 - TRANSFER_YIELD: returns the transfer volume from x neighboring water district based on reservoir levels
 - MCGWB_SWITCH: a switch that turns the capacity of extraction to 0 on May 1st based on reservoir storage
 - Soquel_SWITCH: a switch that turns the supply from Soquel ASR on/off based on reservoir levels- checked between May and September
- File: Python file that imports assumptions from relevant csv files
 - Name: scwsm_model_assumptions.py
 - Note: Csv files imported not included in Github repository
- File: Python file that includes parameter classes for relevant operations and constraints
 - Parameter classes:
 - Max_flow_below_LL: sets the maximum flows downstream of the reservoir based on the month of the year
 - Hydraulic_constraint: calculates the hydraulic loss factor for a pump in the system
 - Note: not included in Github repository
- File: Python file that includes parameter classes for relevant water rights assumptions and operations
 - Parameter classes:
 - Felton_to_LL_Annual_License: calculates the annual license at a diversion within the system

- WATER_RIGHT_NEWELL_TO_GHWTP: returns current value of reservoir license tied to use of upstream source for water supply
 - WATER_RIGHT_NEWELL_DIV_TO_LL: returns current value of license tied to certain diversion
 - PRECIP_WATER_IN_LL: returns the volume of water from local rainfall over the reservoir
 - FELTON_WATER_IN_LL: returns the volume of water in the reservoir originating from a certain source
 - NEWELL_WATER_IN_LL: calculates the volume of water in the reservoir from a specific upstream source
 - USEABLE_PRCP_TO_GHWTP: returns volume available from precipitation in reservoir available for water treatment plant
 - WATER_MONTH_TYPE: returns the water month type based on flow from an upstream source
 - SWITCH_MCASR_INJECTION: returns the binary value based on the water month type parameter
 - SWITCH_TRANSFERS returns the binary value based on the water month type parameter
 - Note: not included in Github repository
- File: Python file that runs the simulations used in the paper
 - Name: Run_simulations_paper_Sherlock.py
 - Note: run on HPC system; you can run it on a local machine, but can comment lines to run different sets of policy simulations. On a local machine, it will take 1-2 minutes per simulation.
- File: Python file that runs the infrastructure financing & rate design sensitivity analysis used in the SI
 - Name: Run_simulations_Sherlock_InfRates_SA.py
 - Note: run on HPC system
- File: Python file that runs the demand & demographics sensitivity analysis used in the SI
 - Name: Run_simulations_Sherlock_DemandDemographics_SA.py
 - Note: run on HPC system
- Subfolder: plots
 - File: Jupyter notebook file to create paper plot Figure 1
 - Name: Figure1_ModelFramework.ipynb
 - File: Python file to process the data for and create paper plot Figure 2
 - Name: Figure2_AddClimate_Sherlock.py
 - Note: this figure involves a lot of data and should be run on an HPC system
 - File: Python file to create paper plot Figure 3
 - Name: Figure3_AddClimate_Sherlock.py
 - Note: this figure involves a lot of data and should be run on an HPC system
 - File: Python file to process and plot Figure 4
 - Name: Figure4_Sherlock.py

- File: Python file to process and plot Figure 5
 - Name: Figure5_Sherlock.py
- File: Python file with functions for processing the different types of data across results figure files
 - Name: processing_functions.py
- File: Jupyter notebook file to plot the timeseries and histograms for Figure S2
 - Name: SI_FigureS2_Timeseries_Infrastructure_Deployment.ipynb
- File: Jupyter notebook file to plot the scatter plots and hypervolumes for the sensitivity analysis results for the optimization (Figures S3, S4)
 - Name: SI_FigS3_FigS4_SA_opt_results.ipynb
- File: Jupyter notebook file to create boxplots for infrastructure financing and rate design sensitivity analysis (Figure S5)
 - Name: SI_SensitivityAnalysis_InfRates.ipynb
- File: Jupyter notebook file to create boxplots for demands and demographics sensitivity analysis (Figure S6)
 - Name: SI_SensitivityAnalysis_DemandDemographics.ipynb
- File: Jupyter notebook file to create the boxplots and histograms for Figures S7, S11
 - Name: SI_FigS7_FigS11_policies_and_boxplots.ipynb
- File: Jupyter notebook file to create the timeseries in Figure S9
 - Name: SI_FigureS9_baseline_infrastructure_policy.ipynb
- File: Jupyter notebook file to create the step plots in Figure S10 comparing infrastructure planning approaches
 - Name: SI_FigureS10_number_of_inf_options.ipynb
- File: Jupyter notebook file to illustrate the ROF dynamics shown in Figures S14, S15
 - Name: SI_FigS14_FigS15_ROF_Dynamics.ipynb
- File: Jupyter notebook file to plot the hypervolume for the baseline optimization results
 - Name: SI_FigureS16_hypervolume.ipynb
- File: R file to plot the correlation between measured and modeled water demands across income groups (Figure S17)
 - Name: SI_FigureS17_Data_Fit_Income_Group.R
- File: Python file to plot a different income estimate for Figure 2 (Figure S18)
- File: Python file to plot a different income estimate for Figure 3 (Figure S19)
- File: Python files (2) to process and plot a different income estimate for Figure 4 (Figure S20)
 - Name(s): SI_FigureS20_Fig4_AR_IE_Sherlock.py
 - Name(s): SI_FigureS20_ProcessData_AR_IncomeEstimate_Sherlock.py
- File: Jupyter notebook file to illustrate the impacts of building infrastructure on water rates within the model (Figure S23)
 - Name: SI_FigureS23_Tiimeseries_Revenue_Rate_Assumptions.ipynb
- File: Python file to compile the statistics for the income estimates in Table S8
 - Name: SI_TableS8_income_estimates_Sherlock.py

- Subfolder: SI_CMIP6
 - File: CalAdapt precipitation data
 - Name: cal_adapt_prec.csv
 - File: CalAdapt temperature data
 - Name: cal_adapt_temp.csv
 - File: CMIP6 precipitation data
 - Name: cmip6_prec.csv
 - File: CMIP6 temperature data
 - Name: cmip6_temp.csv
 - File: Python file for plotting climate anomalies (Figure S1)
 - Name: plot_CMIP6_figure.py

Subfolder: results

- Currently, this subfolder is empty. If you run the model, sample result files include:
 - results... .csv → contain daily results from the water system model in Pywr
 - df_cashflow... .csv → contain monthly results for the cashflow model
 - df_timetracker... .csv → contain the costs, names, and timing for the infrastructure options deployed
 - arr_hh... .npz → contain household-level results